Trigonometrical FS Expansion –

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin(n\omega_0 t)]$$

where $a_0 = \frac{1}{T}\int_T x(t)\cdot dt$ = avg value of $x(t)$

$$a_n = \frac{2}{T}\int_T x(t)\cdot \cos(n\omega_0 t)\, dt$$

$$b_n = \frac{2}{T}\int_T x(t) \sin(n\omega_0 t)\, dt$$

Q: FS expansion of a real signal $x(t)$ is given by—

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{3}{4+(3n\pi)^2} e^{jn\pi t}$$

. Determine—

(1) T_0 (2) value of A_0 in $A_0 \cos(6\pi t)$ & $A_0 \sin(6\pi t)$

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega_0 t}$$

$$c_n = \frac{3}{4+(3n\pi)^2}$$

$$\therefore A_0 = a_6 = 2\,\text{Real}[c_6] = \frac{6}{4+(18\pi)^2}$$ for $A_0 \cos(6\pi t)$

$$x[n] \rightleftharpoons a_k$$

$$x[n] = \sum_{k=\langle N \rangle} a_k \cdot e^{jk\omega_0 n}$$

& $$a_k = \frac{1}{N} \sum_{k<N} x[n] \cdot e^{-jk\omega_0 n}$$

where, $\boxed{N = \frac{2\pi}{\omega_0}}$ → Time period of signal $x[n]$

(2) $x[n] = 1 + \sin\left(\frac{2\pi}{N}\right)n + 3\cos\left(\frac{2\pi}{N}\right)n + \cos\left(\frac{4\pi}{N}n - \frac{\pi}{2}\right)$

$x[n] = \sum_{k<N>} a_k \cdot e^{jk\omega_0 n}$ $\quad \omega_0 = \frac{2\pi}{N}$ $\quad N=4$ $\quad N=8$

$x[n] = 1 + \left(\frac{3}{2} + \frac{1}{2j}\right)e^{j\left(\frac{2\pi}{N}\right)n} + \left(\frac{3}{2} - \frac{1}{2j}\right)e^{-j\left(\frac{2\pi}{N}\right)n}$

$+ \left(\frac{1}{2}e^{j\pi/2}\right)e^{j2\left(\frac{2\pi}{N}\right)n} + \left(\frac{1}{2}e^{-j\pi/2}\right)e^{-j2\left(\frac{2\pi}{N}\right)n}$

$\therefore \quad a_0 = 1$,

(2) Given that -

$$x(t) = \begin{cases} (t+2) & -2 < t < 0 \\ (-2t+2) & 0 < t < 1 \end{cases}$$

$T_0 = 3$

Find C_n?

$$C_n = \frac{1}{T}\int_T x(t)\, e^{-jn\omega_0 t}\, dt$$

$$C_0 = \frac{1}{3}\int_3 x(t)\, dt = \frac{1}{3} \times 3 = 1$$

$\therefore$

$$C_k = \frac{1}{3}\int_{-2}^{1} x(t)\cdot e^{-jk\omega_0 t}\cdot dt$$

$$= \frac{1}{3}\underbrace{\int_{-2}^{0}(t+2)\cdot e^{-jk\omega_0 t}\cdot dt}_{I_1} + \frac{1}{3}\underbrace{\int_{0}^{1}(-2t+2)\cdot e^{-jk\omega_0 t}\cdot dt}_{I_2}$$

$$I_1 = \left\{\frac{2}{-jk\omega_0} - \frac{1}{(-jk\omega_0)^2} + \frac{e^{j2k\omega_0}}{(-jk\omega_0)^2}\right\}$$

$$I_2 = \left\{ \frac{2\,e^{-jK\omega_0}}{(-jK\omega_0)^2} - \frac{2}{(-jK\omega_0)} - \frac{2}{(-jK\omega_0)^2} \right\}$$

$$\therefore C_K = -\frac{1}{3}\left\{ I_1 + I_2 \right\}$$

$$= \frac{3j}{2\pi^2 K^2}\left\{ e^{jK\cdot\frac{2\pi}{3}} \cdot \sin\left(\frac{K\cdot 2\pi}{3}\right) - 2\cdot e^{-jK\pi/3} \cdot \sin\left(\frac{K\pi}{3}\right) \right\}$$

(2)

$$x(t)=\begin{cases}(t+2) & -2<t<-1\\ 1 & -1<t<1\\ (2-t) & 1<t<2\end{cases}$$

$T=6$

$$C_0=\frac{1}{6}\int_{-2}^{2}x(t)\,dt=\frac{1}{2}$$

$$C_K=\frac{1}{6}\left\{\underbrace{\int_{-2}^{-1}(t+2)\cdot e^{-jK\omega_0 t}\cdot dt}_{I_1}+\underbrace{\int_{-1}^{1}e^{-jK\omega_0 t}\cdot dt}_{I_2}+\underbrace{\int_{1}^{2}(2-t)\cdot e^{-jK\omega_0 t}\cdot dt}_{I_3}\right\}$$

$$I_1 = \left\{ \frac{e^{-jk\omega_0}}{(jk\omega_0)} - \frac{e^{jk\omega_0}}{(jk\omega_0)^2} + \frac{e^{j2k\omega_0}}{(jk\omega_0)^2} \right\}$$

$$I_2 = \left\{ -\frac{e^{-jk\omega_0}}{(jk\omega_0)} + \frac{e^{jk\omega_0}}{(jk\omega_0)} \right\}$$

$$I_3 = \left\{ \frac{e^{-j2k\omega_0}}{(-jk\omega_0)^2} + \frac{e^{-jk\omega_0}}{(jk\omega_0)} - \frac{e^{-jk\omega_0}}{(jk\omega_0)^2} \right\}$$